Archita Agrawal

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Education

Indian Institute of Information Technology Kota **Aug 2023 – Present**
B.Tech in Electronics and Communication Engineering ***CGPA: 8.64/10.0***

Technical Skills

CS Fundamentals: Data Structures & Algorithms, OOP, OS, DBMS
Languages: C++, C, Python, JavaScript, TypeScript, SQL
Backend & Databases: Node.js, Express.js, PostgreSQL, REST APIs, Socket.IO, JWT, bcrypt, MongoDB, Mongoose, Zod, Winston
Frontend: React.js, Next.js, Tailwind CSS, GSAP, Framer Motion, Vite, Zustand, React Hook Form, HTML, CSS, Responsive Design
AI Integration: Google Gemini API, LLM Prompt Engineering, Streaming Responses
Tools & Analytics: Git, GitHub, Vercel, Render, Cloudinary, Chart.js, Google Sheets API, Pandas, NumPy

Experience

Gyanama | **Live** **Jan 2026 - June 2026**
Software Engineer Intern

- Built and shipped the production marketing website for Gyanama's AI school operating system, translating features like AI attendance calling and student health scores into a responsive, conversion-focused site with **React 18, TypeScript, and Vite**, styled with **Tailwind CSS** and **shadcn/ui**.
- Implemented scroll-driven animations and interactive product showcases with **Framer Motion**, and integrated a **Cal.com** scheduling widget with an email-form fallback across 9 routed pages.
- Built reusable, accessible UI components and optimized the production build with lazy-loading, code-splitting, and an SEO-tuned build pipeline, improving **Core Web Vitals** and load times across devices.

Samar Trading | **Live** **May 2026 - August 2026**
Freelance Software Developer

- Designed and implemented an automated backend data pipeline integrating the **Google Sheets API** to capture and process client form submissions with **near-zero data loss** under concurrent multi-user load, eliminating a manual workflow entirely; **validated with unit and integration tests** across concurrent user scenarios.
- Built and documented **RESTful API endpoints** with schema-based input validation and structured error handling, reducing integration issues reported by downstream collaborators; decomposed the codebase into **10+ component modules** with strict interface contracts and **separation of concerns** to enable parallel development.

Projects

Mentora AI | *React, Node.js, Socket.IO, MongoDB, Gemini AI* | **Live** **2026**

- Architected a **real-time learn-by-teaching platform** based on the Feynman Technique, where users explain concepts to an AI student that asks the questions a confused learner would.
- Engineered a **low-latency Socket.IO pipeline** delivering word-by-word responses validated server-side for rule violations and repetition, with **JWT auth over httpOnly cookies** across a decoupled REST + WebSocket architecture.
- Built an automated **concept-extraction and scoring engine** on Gemini, implementing **chunked PDF parsing**, retry logic with exponential backoff, and a **three-dimension rubric** whose completeness score is recomputed deterministically from key-point coverage to keep rescores stable—feeding structured gap feedback and **XP/streak rewards**.

PawConnect | *React, Node.js, MongoDB, Express.js, Socket.IO* | **Live** **2026**

- Designed a multi-tier **REST API backend (8 route modules)** with **JWT auth and 4-level RBAC**; applied **OOP principles** for modular, maintainable service architecture with strict separation of concerns.
- Built a **low-latency Socket.IO event pipeline** managing concurrent user sessions, optimized for throughput and minimal event-processing latency under load; **debugged and resolved race conditions** across sessions and integrated **Cloudinary**, **Nominatim geocoding**, and **dual-confirmation transactional workflows** with 4 real-time Chart.js dashboards, enforcing multi-step data integrity with structured error handling and audit-trail patterns.

Heart Disease Prediction | *Random Forest, Feature Engineering* | **NIT Patna 2026** | **GitHub** **2026**

- Achieved **85.87% accuracy** and a **0.9167 ROC-AUC score** by optimizing a **Random Forest classifier** on a 920-record UCI dataset using **Grid Search** and **5-Fold Cross-Validation**; research accepted at a peer-reviewed national conference.
- Reduced model misclassification risk by performing **feature importance analysis** across 12 cardiovascular parameters, eliminating low-signal features and improving prediction interpretability for clinical decision support.

Achievements

- **Algorithmic Problem Solving:** Solved **500+ problems on LeetCode** and **550+ across competitive platforms**, with consistent focus on graphs, dynamic programming, and system-level data structures.
- **Competitive Coding:** Competed in **50+ CodeChef** contests, earning a peak **3-Star** rating; strong background in time-complexity optimization under contest constraints.
- **Research Publication:** Peer-reviewed ML paper on cardiovascular risk prediction **accepted at NIT Patna Conference 2026**.
- **Odoo Hackathon 2026 Finalist:** Top **13%** of 20,363 registrations; in the 24-hour finale, co-built **DealFlow360**, a quote-to-cash platform in **Spring Boot and React** with discount approvals, subscription billing, and warehouse allocation.